

FetchHound: From Human-Canine to Human-Robot Object Search

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I. INTRODUCTION

Consider a home robot searching for objects in an environment contains several visually similar candidates, and the human agent supplies only partial information about the target’s location and appearance. Such a search process engages three capabilities: perception of candidate objects and partner state, action execution with the body and sensors, and attention coordination, which is the choice of where to direct attention next. This project targets the third, which we formulate as the agent’s *social attention policy*: the choice, at each step of the search, of how to allocate attention between candidate objects in the environment and the human partner who holds partial information about the target.

Constructing a social attention policy requires three properties together: a partner representation as a state to be attended to, a temporal structure governing how and when to allocate attention, and an action space tractable enough to be learnable from limited data. Single-agent active perception such as VLFM (1), ESC (2), and the state-machine-based CogNav (3), optimizes attention over the environment alone. The partner does not appear in the policy’s state, even when that policy is given explicit hierarchical structure. Gaze-aware human–robot interaction (4; 5; 6) represents the partner explicitly, but only at the instantaneous level. It detects current gaze direction and reacts, without a temporal allocation between partner-directed and environment-directed sensing. Principled formulations of collaborative search based on POMDPs (7; 8; 9; 10; 11) and theory-of-mind-aware multi-agent search (12) do model the partner and partial observability explicitly. However, scaling these formulations to a temporally extended attention allocation policy that is learnable from the limited interaction data available remains an open problem.

Outside robotics, dogs have collaborated with humans on various attention-coordinated tasks. We specifically focus on locating objects without shared language. Canine cognition research has characterized this coordination as a small set of discrete regimes: gaze following under ostensive cues (13; 14), independent exploration prior to help-seeking (15; 16), gaze alternation for joint reference (17; 18), and check-back at impasse (19; 20), together with quantitative statistics on how these regimes are sequenced.

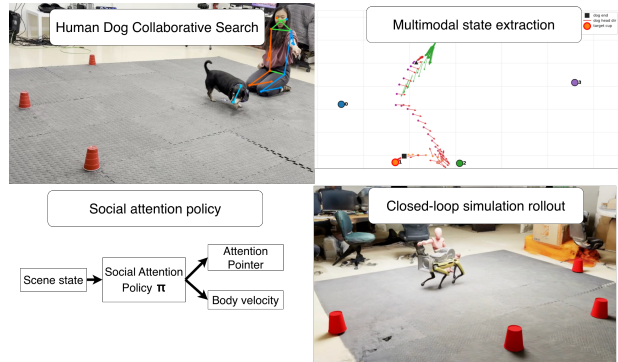


Fig. 1: **System pipeline.** Naturalistic recording of human–dog collaborative search → 3D pose and gaze-proxy estimation → social attention policy learning → closed-loop simulation rollout.

We hypothesize that this protocol can serve as the structural prior for a robot’s social attention policy. The conditions that have produced and stabilized it in dogs match those faced by a mobile robot. The sensing channels available to a dog coincide with those available to a mobile robot. The task itself, locating a target under partial information held by a human partner, is the same. This motivates the central question of this paper: *can attention coordination patterns extracted from human–dog interaction serve as effective priors for a robot’s social attention policy in collaborative object search?*

We aim to address it through three complementary investigations: i) the structure of the coordination, ii) its transfer to a robot policy, and iii) its functional impact on collaborative search performance. Our contributions are threefold: a) the first pipeline to transfer a cognitive-level policy, rather than a motor-level skill, from animal behavior to a robot; b) a method for extracting object-level attention supervision via 3D reconstruction, simulation re-rendering, and head-direction ray-casting; and c) an empirical comparison of the resulting policy against various baselines on collaborative search success and attention pattern similarity.

II. TECHNICAL APPROACH

We formulate collaborative object search with embodied partner cues as a discrete-time sequential decision problem. At each frame t , the state s_t comprises (i) the agent’s own kinematic features (velocity, head offset,

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head direction); (ii) the human partner’s relative pose together with head direction and pointing direction; (iii) the relative positions of K candidate objects; and (iv) a per-entity visibility tier (central, peripheral, or occluded) computed from the agent’s head direction against scene geometry. The action $\mathbf{a}_t = (c_t, \hat{\mathbf{d}}_t, s_t)$ comprises a categorical attention target c_t over the partner, the K candidate objects, and a no-attention class $\perp$; a unit body direction $\hat{\mathbf{d}}_t \in \mathbb{S}^2$; and a non-negative scalar speed $s_t \in \mathbb{R}_{\geq 0}$. The objective is to learn a policy $\pi_\theta(\mathbf{a}_{t+1} \mid \mathbf{s}_{t-T+1:t})$ that predicts the next-frame action from a T -frame history window.

We instantiate the policy via behavior cloning from real human–dog collaborative search demonstrations, leveraging real human dog collaboration behavior as source of attention coordination strategies. To obtain clean per-frame attention labels at scale, we reconstruct each real trial and re-render and augment in simulation, where head-direction ray-casts from a virtual egocentric camera on the dog identify the attended entity, bypassing the noise of physical eye-tracking on animals. As illustrated in Fig. 1, our pipeline extracts this shared per-frame state representation from real and simulated trials, trains the social attention policy by behavior cloning, and evaluates it via closed-loop rollouts in Unity (Fig. 2).

Data: We train on two sources sharing the per-frame state–action representation above: real human–dog collaborative search trials, and synthetic trials generated in simulation with a Boston Dynamics Spot and calibrated to statistics from the real trials. In the real data collection process, a human–dog dyad collaborates on a treat-finding task. All candidate objects are pre-rubbed with treats to suppress olfactory cues. The treat is hidden; and the dog (the searcher) has no information about the target beyond what the human (the instructor) provides. The human instructor is required to give an initial multimodal cue and is allowed to continue prompting the canine searcher throughout the search. The real trials are 3D-reconstructed and re-rendered in a common metric frame, from which both the dog’s gaze proxy and the human’s pointing direction are derived; the synthetic trials augment the limited real pool.

Behavior Cloning Policies: The policy maps a window of past states to the next action through a dual head: a categorical attention head trained with cross-entropy, and a movement head regressing the unit direction and scalar speed. We train three architectures that share this decoder and differ only in how the window is summarized: an *MLP* (mean pool plus most recent frame), an *LSTM* (final hidden state), and a *Transformer* (pooled output tokens). We compare against two hand-crafted baselines: a *nearest-cup* policy that attends to the closest visible candidate, and a *point-follower* policy that tracks the human’s pointing direction while observable

TABLE I: Per-policy comparison across synthetic and real human–dog trials.

Policy	Synthetic		Real	
	attn-acc	success $\uparrow$	attn-acc	success $\uparrow$
nearest_cup	0.21	0.27	0.17	0.33
point_follower	<u>0.43</u>	<u>1.00</u>	0.16	0.40
MLP	0.38	0.75	0.15	0.37
LSTM	0.04	0.63	0.18	<u>0.83</u>
Transformer	0.12	0.87	<u>0.30</u>	0.63

and retains the last cup seen otherwise.

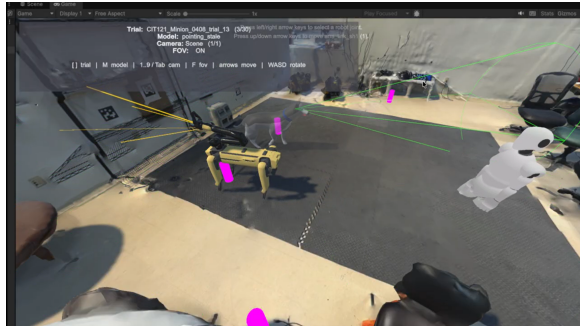


Fig. 2: **Unity visualization.** The yellow Spot robot is the policy rollout, while the transparent dog is the real dog replayed from recorded trajectory data.

III. DISCUSSION & FUTURE DIRECTION

Table I shows that all three learned policies achieve non-trivial task success in both simulation and real trials, demonstrating that a social attention policy can be extracted from human–dog interaction and transferred to a robot. The learned policies outperform the nearest_cup baseline on task success in both domains, confirming that the policy leverages the human’s referential signal rather than a naive geometric heuristic. Additionally, the current baselines, a geometric heuristic and an oracle with direct access to the pointing signal, do not adequately bracket the space of realistic non-learned strategies.

While our pipeline enables learning and validating social attention policies, several limitations remain. The training corpus remains small, likely limiting generalization to novel scene configurations. Our state representation assumes clean 3D reconstructions and ignores noise from depth cameras and off-the-shelf pose estimators, leaving the sim-to-real gap uncharacterized. Our architecture sweep varies only the temporal aggregator (MLP, LSTM, Transformer); distributional decoders such as diffusion policies (21) and hierarchical or belief-tracking architectures remain open directions. More importantly, the current setup is deliberately minimal, we still need to scaling to richer environments with candidate objects, and obstacles will test whether the learned behaviors generalize beyond controlled lab conditions.

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